Social media use in adolescent clinical and non-clinical populations

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Abstract

There is much concern that adolescent mental health declines and concurrent increases in social media use are linked. To address this, previous research has mostly analysed school- or community-based adolescent samples, systematically overlooking those diagnosed with mental health conditions. However, such populations are especially at risk and might have distinct patterns of social media use. In this Registered Report, we will address this oversight by analysing the UK Mental Health of Children and Young People survey (approx. *N* = 4057, 11-19 years old) that collected standardised multi-informant diagnostic mental health assessments, as well as time- and engagement-based metrics of social media use. We will use linear regressions and equivalence testing to test our hypotheses of differences in social media use between adolescents 1) with and without mental health conditions, and 2) with internalising/externalising conditions, compared both with each other and with those adolescents without a mental health condition.

Introduction

Adolescents around the world have experienced a stark decline in their mental health over the past decade[1]. Recent UK data suggests that one in six 7-16-year-olds and one in four 17-19-year-olds have a probable mental health condition, a clear rise from the one in nine and one in ten recorded in 2017, respectively[2]. As 48% of those with a mental health condition first experience relevant symptoms before the age of 18[3], this increased mental health burden will negatively impact society, the economy, as well as adolescent and adult life[4]. Many have raised concerns that this trend has been caused, at least in part, by increased adolescent social media use, which has revolutionised how adolescents live, learn and interact (93% of 12-17-year-olds now report having a social media profile[5]).

To address these concerns, academic investigation of social media use and adolescent mental health has increased substantially in recent years[6]. Research teams have recruited adolescent populations in schools, universities or as part of population-wide samples to identify cross-sectional and longitudinal links between increased smartphone or social media use and scores on questionnaires of depression[7,8], anxiety[9], disordered eating[10] and other mental health symptoms[10–12]. These studies have primarily found small negative relationships. Some researchers have used these to argue that there exists a causal link between social media use and mental health declines (i.e., "screen time, perhaps especially social media, may have larger effects on adolescent girls' mental health than on boys' and that is indeed what we found, with social media significantly correlated with depressive symptoms [...]"[13] p. 13). Such arguments, in turn, have been used to call for restrictive policy regulations to limit smartphone and social media use in adolescent age groups[14].

However, many researchers have also questioned the strength of the current evidence base and highlighted that existing studies do not support the idea that there is a causal relationship linking social media use to mental health. Indeed, the literature provides many conflicting results[15]. Researchers have not only debated about a lack of temporal or causal evidence[16], but have also disagreed about what effect sizes matter[17–19], and how to deal with the substantial individual differences present[20,21], which have been linked to factors such as age[22], gender [23,24], and ethnicity[25].

Across these topics of debate, however, researchers have largely overlooked how the kind of populations being studied limits their ability to draw meaningful inferences about social media's relationship with adolescent mental health in the first place. To date, most studies have examined school- or community-based adolescent samples[15,16,26], relating scores on mental health questionnaires (e.g., the Hospital Anxiety and Depression Scale[9,27]) to time spent on social media. The rationale for doing so is that questionnaires capturing continuous clinical symptomatology in mostly healthy samples are informative when reasoning about mental health in adolescents with clinical

conditions. However, this approach is not a suitable surrogate for studying links between social media and mental health in clinical populations, for two main reasons. First, it reduces the complexity of clinical presentations to the tail end of variation in select mental health symptoms among mostly healthy individuals. Second, it ignores the potentially important differences between those who endorse symptoms on a questionnaire and those who have received a formal mental health diagnosis. For example, an adolescent can score very highly on a questionnaire measuring depressive symptoms, but not meet criteria for a diagnosis if the queried symptoms have only been present for a short time, or if they are better explained by a different condition or situation.

To address this issue, select studies have moved beyond such a basic approach and dichotomised symptom severity by applying cut-off scores to mental health questionnaires to reflect the presence or absence of a mental health diagnosis[23]. However, dichotomisation does not solve many of the problems highlighted above and is known to have low sensitivity when predicting clinical diagnoses. Indeed, those with clinically significant symptoms can score below the threshold[28,29]. In other words, although researchers would like to demonstrate the presence or absence of specific links between mental health symptoms and social media use, the samples they study might not be appropriate for these goals.

Importantly, the assumption that links between social media use and mental health found in community samples will generalise to clinical populations has not yet been systematically tested. To our knowledge, only a few studies - most of which qualitative - have documented different social media use experiences in clinical adolescent populations, including those fulfilling stringent diagnostic criteria for a clinical condition, attending mental health services or being hospitalised for suicidal ideation and suicide attempts[30–33]. Adolescents in these studies reported both positive and negative social media experiences, such as enhanced social connection and trouble downregulating their use. Broadly speaking, these experiences aligned with established risk and protective factors previously linked to mental health in offline spaces and suggest there is no clear-cut positive or negative association between mental health and social media use. The studies also raise the idea that vulnerable youth might experience heightened emotional responses to social media use. However, this has not been directly assessed due to a lack of non-clinical comparison groups. Studies comparing clinical and non-clinical samples are necessary to understand whether specific dimensions of social media use differ between clinical and non-clinical populations.

Due to the lack of research on clinical populations[34], the important question of whether social media might be differentially related to mental health across diverse adolescent clinical groups also remains unaddressed. For example, an adolescent with an internalising condition (e.g., generalised anxiety

disorder or depressive disorder) might use and feel impacted by social media differently than an adolescent with an externalising condition (e.g., attention deficit hyperactivity disorder or conduct disorder). This is because, despite both groups being diagnosed with a clinical condition, their experiences of psychopathology are qualitatively different. Internalising conditions involve negative emotionality towards the self, expressed through ruminative thought patterns, worries and social withdrawal[35]. On the contrary, externalising conditions involve negative emotionality towards others, expressed through impulsivity, risk-taking, and disinhibition[36]. Studies that only examine community samples cannot comprehensively account for and investigate such clinical diversity. This is a substantial oversight, given the need for research to understand how social media use relates to the growing number of adolescents experiencing mental health symptoms at clinical levels.

This Registered Report will provide critical data and evidence-based insights into how social media and mental health are related across clinical and non-clinical adolescent populations (see Table 1 for an overview). We note that, due to the cross-sectional nature of the data and planned analyses, the study results will not provide causal evidence. Hence, all reported coefficients indicate associations, with the possibility of bidirectional relationships and third variables affecting social media use, mental health or the relationship between the two. We will analyse the representative Mental Health of Children and Young People (MHCYP) study[37], a cross-sectional survey carried out by NHS Digital in 2017 that collected data from over 4,000 adolescents (11-19 years old) in England. In place of completing self-report measures of mental health, the participants in this study underwent multi-informant standardised diagnostic assessments evaluated by professional clinical raters for different mental health diagnoses (Table 2).

Further, to gain a comprehensive understanding of how social media use differs across clinical and non-clinical adolescent populations, we will examine both quantitative and qualitative dimensions of social media use. It has long been established that measuring only time spent on social media provides a crude and simplistic estimate of social media use, conflating distinct analytical levels and missing a rich range of psychological factors such as appraisal and motivations that might vary as a function of mental health[38]. Researchers have therefore called for quantitative time-based measures of social media use to be complemented by more qualitative engagement-based measures capturing adolescents' social media activities and their appraisal of them[16,39–43]. Such practice is, however, still relatively rare. In this Registered Report, we will include both types of measures, measuring time spent on social media and dimensions of social media engagement that could incur mental health risks (i.e., online social comparison, monitoring and impact of online feedback, lack of control over time spent online) and benefits (i.e., online friendship, as well as opportunities for honest self-disclosure and authentic self-presentation). By complementing quantitative and qualitative measures of social media

use, we will provide a more solid foundation for mechanistic research aimed at informing future targeted interventions, clinical practice and policy actions benefiting adolescent mental health.

We will use existing literature on adolescents' mental health in relation to both online and offline contexts (Table 3) to guide our hypotheses and analyses of the data along three lines of inquiry. Specifically, we will evaluate whether social media use differs in adolescents with vs without a mental health condition (Question 1), with an internalising or externalising condition vs without a condition (Question 2), and with an internalising vs externalising condition (Question 3).

First, we expect adolescents with a mental health condition to report engaging with social media differently than those without a condition. For instance, gathering information about peers is particularly important during adolescence when young people develop a sense of personal and social identity[44]. However, high levels of social comparisons (i.e., comparisons with those believed to be of higher status than the self) have been associated with poor mental health[45–49]. Previous work suggests that most social comparisons made on social media sites are upward rather than downward[50], possibly because individuals tend to portray themselves in an ideal manner online[51,52]. Social media could further amplify these processes as platforms offer continuous and more concrete opportunities for comparing oneself to others, such as browsing profiles without initiating social interaction[53]. Indeed, engagement in self-denigrating online social comparisons was a common theme raised by adolescent psychiatric inpatients during qualitative interviews[31]. We therefore expect adolescents with a clinical condition to engage in more online social comparison than those without a condition (Hypothesis H1.1b). In a similar vein (see Table 3 for a detailed overview of the literature in support of each hypothesis), we expect them to spend more time on social media[8,15] (H1.1a), report more lack of control over time spent online[31,54] (H1.1c), while also monitoring[55,56] (H1.1d) and feeling more impacted by online feedback[57,58] (H1.1e).

In contrast, we expect adolescents with a clinical condition to engage less with social media in ways that might protect their mental health. For instance, we hypothesise that adolescents with a mental health condition will be less happy with the number of friends they have online than adolescents without a condition (H1.2f). This is because social connections protect against long-term adverse physical and emotional outcomes, particularly during adolescence[48], and young people with mental health conditions often report difficulties with peers, having few friends and wanting to be alone[59–61]. Similarly, we would expect adolescents with a mental health condition to engage in less honest online self-disclosure[62,63] (H1.2g) and authentic online self-presentation[64,65] (H1.2h) compared to adolescents without a condition (see Table 3).

Second, we predict social media use to vary between clinical groups with different symptomatology, focusing specifically on how the social media use of adolescents with an internalising or externalising condition differs from those without a condition. We know from the mental health literature that internalising conditions involve negative self-views, rumination, worries and social withdrawal[35,66]. Notably, these symptoms are all relevant to how young people present themselves and engage with others on social media[31,32,67]. For instance, adolescents with internalising conditions are more likely to notice discrepancies between their ideal and actual selves[68] and may compensate for these discrepancies via impression management. We expect this process to also occur online, given the multiple affordances social media platforms offer to curate one's image (e.g., deleting old posts, editing new posts). Hence, we hypothesise that adolescents with internalising conditions will be less likely to engage in authentic self-presentation than adolescents without a mental health condition[69–71] (H2.2h). Further (see Table 3), we hypothesise that adolescents with internalising conditions will spend more time on social media[8,15,72] (H2.1a), engage in more online social comparison[30] (H2.1b) and online feedback monitoring [55] (H2.1d), feel more impacted by online feedback[57] (H2.1e), be less happy about online friends (H2.2f) and engage in less authentic self-disclosure[63] (H2.2g) than those without a condition. In contrast, we do not expect a difference in whether they perceive a lack of control over the time they spend online (H2.0c)[54] .

Compared to internalising conditions, externalising conditions involve impulsivity, low self-monitoring, and risk-taking[35,66]. These symptoms will likely be reflected in how social media is used by these clinical groups. Hence, we hypothesise that adolescents with externalising conditions will be more likely than adolescents without a condition to spend more time on social media[73] (H2.3a) and perceive they lack control over the time they spend online[54] (H2.3c). Further, we expect them to also be more dissatisfied with their number of online friends[74] (H2.4f). In contrast, we do not expect differences in the other dimensions of social media engagement (H2.0b, d, e, g, h; see Table 3), since they primarily relate to symptoms of internalising rather than externalising diagnoses.

Our third question examines whether adolescents with internalising conditions use social media differently than those with externalising conditions. Specifically, we expect adolescents with internalising conditions to report engaging in more online social comparison[30] (H3.1b) and online feedback monitoring[56] (H3.1d), feel more impacted by online feedback[57] (H3.1e) and report less self-disclosure[62] (H3.2g) and authentic self-presentation on social media[70] (H3.2h) than those with externalising conditions. In contrast, we expect adolescents with externalising conditions to report having less control over the time they spend online[54,74] compared to adolescents with internalising diagnoses (H3.2c). Further, since adolescents with internalising conditions tend to be unhappy with their social status[75], and adolescents with externalising conditions tend to have trouble making and

keeping friends in both online and offline contexts[74,76], we expect both groups to be equally dissatisfied with their number of online friends (H3.0f). Similarly, we hypothesise that both groups will not differ in the amount of time they spend on social media (H3.0a), given that both adolescents with internalising and externalising symptoms have been reported to spend more time on social media than adolescents without these symptoms[33,77].

Altogether, this study will comprehensively map and compare the relationship between different dimensions of social media use and mental health in both clinical and non-clinical adolescent populations. Hence, it will lay the foundation for future mechanistic and translational research studying which specific social media dimensions relate to mental health in different adolescent groups. This will be crucial to inform translational research, clinical practice, as well as the design of targeted interventions and policies to improve children and adolescents' mental health.

Methods

Ethics information

The MHCYP 2017 survey was reviewed and approved by the West London & GTAC Research Ethics Committee (REC reference: 16/LO/0155) and the Health Research Authority Confidentiality Advisory Group (CAG reference: 16/CAG/0016) in 2016. Both parents and children provided consent to take part in data collection and were compensated with a 10£ voucher for their time. Parents of children under 16 years were interviewed first, and permission was sought to interview their child afterwards; the child then provided assent. Conversely, 17–19-year-olds were directly asked for their consent, with permission subsequently sought for their parents to be interviewed. Access to the data was granted to the research team by NHS Digital (DARS-NIC-424336-T7K7T-v0.6 r).

Design

The MHCYP study is one of a series of national surveys on the mental health of children and young people in England administered in 1999, 2004, 2017, 2021 and 2022. In this Registered Report, we analyse the 2017 wave collected between January and October 2017: the most recent wave to be made available to researchers as well as the first wave to collect comprehensive data on adolescents' social media use and to include 17 to 19-year-olds. We will only analyse data from adolescents aged 11-19 years, a total of approximately 4,057 participants (50.1% male, 49.9% female) out of the full sample of 9,117. The survey was collected using a stratified probability sample of children and young people living in England who were registered with a general practitioner[37]. Data was collected via face-to-face interviews with adolescents and their parents. At the same time, if the family agreed,

questionnaires were mailed to teachers (for the available data and key demographics see MHCYP 2017[37,78]).

Measures

Time spent on social media. We will measure time spent on social media using two questionnaire items: "When you use social media sites or apps, how much time do you spend using them on a typical school day?" (SMTimeSpentS), and "When you use social media sites or apps how much time in total do you spend using them on a typical weekend or holiday day?" (SMTimeSpentW). Participants answered both questions on a 9-point Likert scale: 1 = Less than 30 minutes; 2 = More than 30 minutes but less than an hour; 3 = One to two hours; 4 = Two to three hours; 5 = Three to four hours; 6 = Four to five hours; 7 = Five to six hours; 8 = Six to seven hours; 9 = More than seven hours. A single variable reflecting average social media hours will be created from these two variables (SMTimeSpent, see Supplementary Table 1 for more details). To do so, we will first calculate the mean time in hours for each response. For example, if a participant responded "one to two hours", we will recode this as one and a half hours. Participants that responded "more than seven hours" will be recoded to seven and a half hours, while participants that responded "less than 30 minutes" will be recorded to fifteen minutes (i.e., 0.25 hours). Weekday hours will then be multiplied by 5, and weekend hours will be multiplied by 2, and the products will be summed and divided by 7 to establish a daily mean social media use variable, measured in hours. The SMTimeSpent variable will be coded as continuous[79]. We will use a sensitivity analysis to test whether examining social media use on weekdays vs weekends/holidays separately, rather than as a weighted average, changes the main results.

In the survey, the questions regarding time spent on social media for both weekdays (SMTimeSpentS) and weekends (SMTimeSpentw) were only asked of adolescents who responded that they use social media sites daily or on most days on a previous questionnaire item (SMFreqofUse; 1 = Daily or most days). Hence, participants that reported a lower frequency of social media use (i.e., SMFreqofUse; 2 = A few times a week; 3 = Once a week; 4 = A few times a month; 5 = Once a month; 6 = Less often than once a month) were not asked these questions. To handle the resulting missing data in the SMTimeSpent variable, we will code any adolescents who responded to the SMFreqofUse question that they use social media between "a few times a week" and "once a week" to fifteen minutes (0.25 hours) and adolescents that responded "a few times a month" and "less often than once a month" to zero hours on the SMTimeSpent question (for more information about our approach to missing data see Supplementary Table 2).

241 *Social media engagement.* We will analyse seven qualitative dimensions of social media engagement
242 measured with questionnaire items tapping into experiences associated with both risks and benefits
243 to adolescent mental health. All measures (summarised with related literature in Table 3) were
244 developed in consultation with a young person advisory group (see Supplementary Methods for more
245 information), where children and adolescents defined the dimensions of social media use most
246 relevant to them. The measures related to mental health risks encompass *online social comparison*
247 (i.e., "I compare myself to others on social media"), *lack of control over time spent online* (i.e., "I spend
248 more time on social media than I mean to"), *monitoring of online feedback* (i.e., "I monitor the amount
249 of likes, comments and shares I get on social media") and *the impact of online feedback* (i.e., "The
250 amount of likes, comments and shares I get on social media has an impact on my mood"). On the
251 contrary, the measures indicative of mental health benefits include *online friendship* (i.e., "I am happy
252 with the number of friends I have on social media"), *honest self-disclosure* (i.e., "I can be honest with
253 people on social media sites and apps about how I am feeling) and *authentic self-presentation* (i.e.,
254 "My social media profile is a true reflection of myself").

255 Participants responded to these measures using a 6-point Likert scale (1 = Disagree a lot; 2 = Disagree
256 a little; 3 = Neither agree nor disagree; 4 = Agree a little; 5 = Agree a lot; 6 = Don't know). We will omit
257 the "don't know" responses and code 1-5 responses as continuous, given research suggesting that 5-
258 point continuous classifications perform as well as or occasionally better than categorical
259 classifications[79].

260 *Mental health diagnoses.* Face-to-face interviewers completed the Development and Wellbeing
261 Assessment (DAWBA)[80] with parents and adolescents aged 11 or over to establish mental health
262 diagnoses. During the interview, participants were first led through the 25 item Strengths and
263 Difficulties Questionnaire (SDQ[81]; Supplementary Table 4). Second, the interviewer administered the
264 Development and Wellbeing Assessment (DAWBA), a diagnostic tool shown to have good validity[82]
265 and reliability[83]. The DAWBA uses structured and semi-structured questions to assess the presence
266 and severity of symptoms for a wide range of DSM-5 or ICD-10 mental health disorders (Table 2). Each
267 module starts with a few screening items, which, if answered negatively (indicative of the lack of
268 symptoms), allows the interview to proceed to the next module with no loss of accuracy[82]. There is
269 one exception: if a participant scored highly on the SDQ, the interviewer is directed to ask in depth
270 internalising disorder DAWBA modules, even if participants screened negative on the initial questions.

271 After the initial screening items, the subsequent structured items in the DAWBA modules relate
272 directly to diagnostic criteria in DSM-5 and ICD-10. They are close-ended questions about specific
273 mental health symptoms (e.g., "In the last 4 weeks, have there been times when you have been very

274 sad, miserable, unhappy or tearful?"). If a participant responds positively to these structured items,
275 they are subsequently asked open-ended questions about these problems (e.g., "Please describe your
276 mood - sadness or irritability - and your level of interest in things"). During the assessment,
277 interviewers transcribed open-ended responses verbatim and were also able to add personal
278 comments beneath each response.

279 A team of clinical raters assessed the DAWBA's structured and qualitative information from all
280 informants to decide whether an adolescent showed evidence of a DSM-5 or ICD-10 mental health
281 disorder that would warrant clinical treatment (Supplementary Figure 1 and 2). In order to determine
282 the presence of a diagnosis, clinical raters 1) checked that the answers to structured comments were
283 understood by the participants accurately; 2) interpreted any conflicts between child, parent and
284 teacher responses and decided which assessment to prioritise; 3) identified clinically impairing
285 disorders that did not perfectly fit current operationalised diagnostic criteria or "not otherwise
286 specified diagnosis" such as "other anxiety disorder".

287 Overall, we will code information on mental health diagnoses into two separate variables: 1) a binary
288 variable indicating the presence of any mental health condition (*diagnosis, no diagnosis)* and 2) a
289 categorical variable for the type of diagnosis (*internalising diagnosis*, *externalising diagnosis, no*
290 *diagnosis*). We subdivided diagnoses identified via the diagnostic assessment into internalising and
291 externalising using existing classifications of psychiatric diagnoses[35,66] (see Table 2). This distinction
292 draws from transdiagnostic research showing that different conditions (e.g., anxiety, depression,
293 eating disorders) are often comorbid, share underlying core symptoms[84] and can therefore be grouped
294 to reflect clinical presentations with more validity[35]. Further, given that we have enough power, we
295 will run additional exploratory analyses (see Supplementary Methods) examining responses to all
296 social media questions (a-g) separately for individual diagnoses (e.g., major depressive disorder) or
297 diagnostic categories (e.g., depressive disorders, anxiety disorders).

298 Comorbidity data will be coded into two separate variables: 1) a binary variable indicating within-
299 group comorbidity (any two diagnoses of either internalising *or* externalising: *yes* or *no*; for descriptive
300 purposes only) and 2) a binary variable indicating internalising-externalising between-group
301 comorbidity (any comorbid internalising *and* externalising diagnoses: *yes* or *no).* Individuals who show
302 between-group comorbidity will be removed prior to the analysis of Questions 2 and 3, given our goal
303 to compare social media use between these clinical groups. To increase the clinical utility of this work,
304 we will run a sensitivity analysis for Question 2, including people with between-group comorbidity.

305 **Analysis Plan**

306 We will conduct all statistical analyses in R Version 4.3.1 (R Core Team, 2021), testing the association
307 between time- and engagement-based measures of social media use and mental health diagnoses
308 using equivalence tests and linear regression models[85,86]. To account for multiple tests, we will provide
309 false discovery rate corrected *p*-values for all models[87]. Our analytical approach is based on
310 regressions rather than ANOVAs as the former allow for more diverse predictors, unbalanced groups
311 and inclusion of covariates in potential exploratory analyses[88]. Below, we describe the statistical
312 analyses we will use to test our questions and hypotheses (see Table 1 for more details). Further, the
313 simulated data and analysis code is available on OSF
314 (https://osf.io/s2dwu/?view_only=5acced2ddb884f9481d439a7746f4dd1).

315 ***Questions and Hypotheses***

316 *Question 1: Investigate whether adolescents with any mental health condition use social media*
317 *differently than those without a condition.*

318 To address Question 1, we will test the association between social media use and mental health
319 diagnosis. Specifically, we will estimate linear regression models with mental health diagnosis as a
320 binary predictor (two levels: *diagnosis* vs *no diagnosis*) and social media use as a continuous outcome.
321 The *no diagnosis* group will be set as the reference level for these analyses. Below, we report our null
322 and directional hypotheses, first for dimensions of social media use expected to reflect mental health
323 risks and second, for dimensions expected to reflect mental health benefits in clinical vs non-clinical
324 groups.

325 H0 (1.0) = Adolescents with any mental health diagnosis will not differ from adolescents
326 without a diagnosis in a) time spent on social media, b) online social comparison, c) lacking
327 control over time spent online, d) monitoring of online feedback, e) feeling impacted by online
328 feedback, f) happiness about the number of online friendships, g) honest online self-disclosure
329 and h) authentic self-presentation online.

330 H1 (1.1) = Adolescents with any mental health diagnosis will score higher than adolescents
331 without a diagnosis in a) time on social media, b) online social comparison, c) lacking control
332 over time spent online, d) monitoring of online feedback and e) feeling impacted by online
333 feedback.

334 H1 (1.2) = Adolescents with any mental health diagnosis will score lower than adolescents
335 without a diagnosis in f) happiness about the number of online friendships, g) honest online
336 self-disclosure and h) authentic self-presentation online.

To examine whether social media use varies with mental health severity, we will conduct sensitivity analyses to assess whether the presence of multiple diagnoses is related to the outcomes of interest (a-g). First, we will plot responses to each social media question by the number of diagnoses. Next, we will test for a linear effect of the number of diagnoses on social media question response using linear regression models.

Question 2: Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.

After assessing differences in social media use in adolescents with vs without a mental health condition, we will examine whether adolescents with internalising or externalising conditions use social media differently than adolescents without a condition. Hence, we will conduct linear regression models with diagnostic category as a categorical predictor (three levels: *internalising diagnosis*, *externalising diagnosis, no diagnosis*) and social media use as a continuous outcome. To test our hypotheses, we will examine comparisons between two levels of the *diagnostic category* variable, with the *no diagnosis* set as the reference level. For hypotheses 2.0[i], 2.1 and 2.2, we will report regression coefficients for *internalising* vs *no diagnosis*, while for hypotheses 2.0[e], 2.3 and 2.4 we will report coefficients for *externalising* vs *no diagnosis*. The null hypotheses marked by "e" or "i" indicate that, for the considered comparison, the null hypothesis is our primary hypothesis. Hence, for those dimensions of social media use, we expect no difference between adolescents with internalising or externalising diagnoses and those without a diagnosis. We used "e" to indicate primary null hypotheses related to externalising vs no diagnosis and "i" to indicate our primary null hypotheses related to internalising diagnosis.

H0 (2.0) = Adolescents with internalising or externalising diagnoses will not differ from adolescents without a diagnosis in a) time on social media, b) online social comparison[e], c) lacking control over time spent online[i], d) monitoring of online feedback[e], e) feeling impacted by online feedback[e], f) happiness about the number of online friendships, g) honest online self-disclosure[e] and h) authentic self-presentation online[e].

H1 (2.1) = Adolescents with internalising diagnoses will score higher than adolescents without a diagnosis in a) time on social media, b) online social comparison, d) monitoring of online feedback and e) feeling impacted by online feedback.

H1 (2.2) = Adolescents with internalising diagnoses will score lower than adolescents without a diagnosis in f) happiness about the number of online friendships, g) honest online self-disclosure and h) authentic self-presentation online.

H1 (2.3) = Adolescents with externalising diagnoses will score higher than adolescents without a diagnosis in a) time spent on social media and c) lack of control over time spent online.

H1 (2.4) = Adolescents with externalising diagnoses will score lower than adolescents without a diagnosis in f) happiness about the number of online friendships.

Question 3: Investigate whether adolescents with an internalising mental health condition use social media differently than those with an externalising condition.

To address Question 3, we will examine how adolescents with internalising conditions differ in social media engagement compared to adolescents with externalising conditions. To this aim, we will compare the *internalising* and *externalising* levels of the *diagnostic category* variable described in Question 2, with *internalising* as the reference variable. Also in this case, the null hypotheses marked by "2" indicate our primary hypotheses. Hence, for those dimensions of social media use, we expect no difference between adolescents with internalising and externalising diagnoses.

H0 (3.0) = Adolescents with internalising diagnoses will not differ from adolescents with externalising diagnoses in a) time on social media, b) online social comparison, c) lacking control over time spent online, d) monitoring of online feedback, e) feeling impacted by online feedback, f) happiness about the number of online friendships, g) online self-disclosure and h) authentic self-presentation online.

H1 (3.1) = Adolescents with internalising diagnoses will score higher than adolescents with externalising diagnoses in b) online social comparison, d) monitoring of online feedback and e) feeling impacted by online feedback.

H1 (3.2) = Adolescents with internalising diagnoses will score lower than adolescents with externalising diagnoses in c) lack of control over time spent online, g) online self-disclosure and h) authentic self-presentation online.

For all regression analyses, we will treat the eight dimensions of social media use, including both time- and engagement-based measures, as separate outcomes predicted by information on mental health diagnoses as the only regressor. While it is common in research to use statistical control to remove confounding effects from a regression coefficient, appropriate control variables should be identified only after justifying a causal structure that includes the outcome, predictors, and all theorised confounders. When the selected control variables are inappropriate or remain unjustified, controlling can result in biased regression estimates[89]. Further, recent literature warns against controlling for

399 demographic factors such as sex without thought and instead prompts researchers to interrogate how
400 this variable intersects with the predictors and outcomes under investigation[90]. In the present work,
401 treating sex or age as a confounding variable would mean ignoring the possibility that there are
402 meaningful sex or age differences in the examined relationships. As our goal is to investigate the
403 overall association between social media use and mental health diagnosis, we will provide a
404 descriptive account of the age and sex of adolescents included in each tested model rather than
405 control for these demographics. For example, in Question 1, age and sex of adolescents with diagnosis
406 vs without a diagnosis will be reported for descriptive purposes.

407 ***Equivalence testing***

408 Given that null hypothesis significance testing does not allow a comprehensive interpretation of
409 statistically non-significant results[91], each model will be complemented by equivalence tests[91] to
410 quantify support for the null hypothesis (H0, 1-3). This involves assessing whether the 90% confidence
411 intervals for the effect size lie inside pre-specified equivalence bounds that indicate the smallest effect
412 size of interest. If the confidence intervals lie inside the equivalence bounds, the effect size is
413 interpreted as negligible. On the contrary, if the confidence intervals spread outside the equivalence
414 bounds, the effect size is considered meaningful in size. In this case, the 90% confidence intervals are
415 used rather than 95% because the effect size is tested against two equivalence bounds separately (i.e.,
416 the upper and lower bound), reflecting $(1–2\alpha) \times 100\%$ (see Table 1 for more details).

417 We established the equivalence bounds by identifying a theoretically meaningful SESOI (d = 0.4, refer
418 to Supplementary Methods for a detailed explanation). After an extensive scoping exercise to identify
419 a suitable theoretical foundation, we determined that the most relevant benchmark for our research
420 questions are everyday behaviours linked to mental health, such as sleep and physical activity. These
421 behaviours, much like social media use, are a regular part of daily routines but, unlike the latter, they
422 are well-established markers of psychopathology, based on both theory [92–94] and matched empirical
423 evidence [95,96]. Consequently, if the actual effect size of social media use is comparable to that observed
424 for behaviours like sleep and physical activity, we can confidently conclude that social media use also
425 represents an everyday behaviour that exhibits meaningful group-level differences between clinical
426 and non-clinical populations. The interpretation and analysis plan are presented separately for each
427 hypothesis in Table 1.

428 Overall, we will infer support for the null hypotheses (i.e., no meaningful difference in social media
429 use between groups) if the 90% confidence interval for this association lies within the equivalence
430 bounds. Of note, while a theoretical SESOI based on everyday behaviours and their link to mental

431 health will serve as our primary effect size of interest to allow for a clear confirmatory approach, we
432 also identified secondary SESOIs based on the effect sizes that are practically and clinically meaningful.
433 These secondary SESOIs will play an important role in supporting the interpretation of our findings (as
434 detailed in Supplementary Methods), ensuring the applicability of our study to both academic and
435 practical domains.

436 Overall, we will infer support for the alternative hypotheses if 1) the coefficient for the association of
437 social media use and our grouping variable for mental health diagnosis is significant, 2) the association
438 follows the hypothesised direction and 3) the CIs for the association do not fall within the equivalence
439 bounds. The interpretation and analysis plan are presented separately for each hypothesis in Table 1.
440 Further, we detail the planned exploratory analyses in Supplementary Methods.

441 ***Sampling plan***

442 *Inclusion criteria and sample size*

443 We will include all participants in the MHCYP 2017 survey aged between 11 and 19 years at the time
444 of assessment who have answered any of the social media use questions with anything other than
445 "don't know". Only individuals with diagnostic information, including the absence of a diagnosis, are
446 included in the MHCYP dataset. Since we will examine each social media use question separately, we
447 will include participants that answered at least one of the seven engagement questions or the
448 question about time spent on social media. No specific documentation is available regarding the
449 response rates for the social media questions. Given that the questionnaires were administered
450 through interviews with professionals, we expect minimal missing data for the social media
451 responses[97]. Hence, using a conservative estimate, we will assume a 5% missingness for our power
452 calculations.

453 To address Question 1, all adolescents with and without a diagnosis will be included in our analysis.
454 Hence, in Question 1, we will include a) participants with conditions other than internalising or
455 externalising disorders (i.e., autism spectrum disorder, tic disorder and psychotic disorders; see Table
456 2); b) participants with within-group comorbidity (multiple internalising *or* externalising diagnoses; for
457 example comorbid depressive disorder and social anxiety disorder); c) participants with between-
458 group comorbidity (both internalising *and* externalising conditions; for example, comorbid depressive
459 disorder and attention deficit hyperactivity disorder).

460 In contrast, for Questions 2 and 3, we will include a) participants with internalising or externalising
461 conditions only; and b) participants with within-group comorbidity (multiple internalising or

externalising diagnoses). Hence, participants with other conditions or between-group comorbidity will be excluded from the main analysis, given our goal to compare social media use in adolescents with externalising and internalising conditions. However, we will run sensitivity analyses to test the impact of including participants with within-group comorbidities in Questions 2 and 3, and between-group comorbidities in Question 2.

Based on summary demographics[78], after accounting for potential missing data, we estimate the sample of complete cases to be around *N*=3,854 (accounting for 5% missingness in the sample of *N*=4,057, 11-19-year-olds), approximately 15% of whom received at least one mental health diagnosis (*N*=577). Available documentation[78] suggests that approximately 19.2% of individuals with internalizing conditions have at least one comorbid externalizing condition, and approximately 28% of individuals with externalizing diagnoses have at least one comorbid internalizing diagnosis. This suggests approximately *N*=370 with internalising-only diagnoses and *N*=199 with externalising-only diagnoses.

Power calculations

We calculated power by setting the smallest effect size of interest (SESOI, *d* = .4, refer to Supplementary Methods for a detailed explanation), the alpha level to 0.05 and the estimated sample size. Regarding sample size, we estimated a total sample size of *N* = 3,854 based on existing MHCYP documentation. For the equivalence tests, power was calculated using the TOSTER package in R (Caldwell, 2022; Lakens, 2018). For the regression models, power was determined using the pwr package (Champely, S., 2020) in R. The code for these calculations is available on the OSF (see https://osf.io/s2dwu/?view_only=5acced2ddb884f9481d439a7746f4dd1).

H0 (1). We calculated the exact power for equivalence testing to detect a SESOI of d = 0.4 given at least N≥577 individuals with a diagnosis and N≥3,277 individuals without a diagnosis. Results indicate 100% power to reject the presence of effects that are larger than d = 0.4.

H1 (1). We calculated power to detect a statistical effect of a diagnosis (yes/no) on social media responses using linear regression. Results indicate 100% power to detect the SESOI (d=0.4) with at least N≥577 individuals with a diagnosis and N≥3,277 individuals without a diagnosis.

H0 (2). We calculated the exact power for equivalence testing to detect a SESOI of d = 0.4 given at least N≥370 individuals with an internalising only diagnosis and N≥3,277 with no

diagnosis. Results indicate 100% power to reject the presence of effects that are larger than d = 0.4.

H1 (2). We calculated power to detect a statistical effect of internalising diagnosis type (i.e., internalising vs no diagnosis) on social media responses using linear regression. Results indicate 100% power to detect the SESOI (d=0.4) with at least N≥370 individuals with an internalising-only diagnosis and N≥3,277 with no diagnosis. We calculated power to detect an effect of externalising diagnosis type (i.e., externalising vs no diagnosis) on social media engagement responses using linear regression. Results indicate 100% power to detect the SESOI (d=0.4) with at least N≥199 individuals with an externalising-only diagnosis and N≥3,277 with no diagnosis.

H0 (3). We calculated the exact power for equivalence testing to detect a SESOI of d = 0.4 given at least N≥370 individuals with internalising only and N≥199 with externalising only diagnoses. Results indicate 96% power to reject the presence of effects that are larger than d = 0.4.

H1 (3). We calculated the power to detect a statistical effect of internalising only vs externalising only diagnosis type on social media responses. Results indicate 99.8% power to the SESOI (d=0.4) with at least N≥370 individuals with internalising only and N≥199 with externalising only diagnoses.

In addition to our a-priori power calculations, we will conduct power sensitivity analyses once our final sample size is known. That is, we will determine the smallest effect size that is observable with 95% power, given an alpha of 0.05 and our final sample size.

Data availability

The MHCYP dataset is held on behalf of NHS Digital by the UK Data Service. Restrictions apply to the availability of this data, which were used under license for this study. Data access can be requested by applying to the Data Access Request Service (DARS). Researchers interested in accessing the data can find further information on the DARS website (I.e., https://digital.nhs.uk/services/data-access-request-service-dars/dars-guidance).

Code availability

The code used to run the power analysis scripts has been made available on the OSF (https://osf.io/s2dwu/?view_only=5acced2ddb884f9481d439a7746f4dd1). On acceptance of the Stage 2 manuscript, we will make the code of all analyses available in R Markdown format on the Open Science Framework.

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Author contributions

We present author contributions according to the CRediT (Contributor Roles Taxonomy).

L.F: Conceptualisation, methodology, formal analysis, writing - original draft, writing - review & editing.

A.F: Conceptualisation, methodology, formal analysis, writing - review & editing.

A.P: methodology, writing - review & editing, supervision.

T.F: writing - review & editing, supervision, resources, project administration.

A.O: Conceptualisation, methodology, writing - original draft, writing - review & editing, supervision.

Of note, none of the authors that conceptualised the analysis previously accessed the MHCYP dataset.

For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

Competing interests

The authors declare no competing interests.

Tables

Table 1. Design Table summarising the study's research questions, hypotheses, power calculations, analyses, and conditions for interpretation.

Question	Hypothesis	Power analysis	Analysis plan	Interpretation given to different outcomes
1) Investigate whether adolescents with any mental health condition use social media differently than those without a condition.	Hypothesis 1.1) Adolescents with any mental health diagnosis will score higher than adolescents without a diagnosis in a) time spent on social media, b) online social comparison, c) lack of control over time spent online, d) monitoring of online feedback and e) feeling impacted by online feedback.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least N≥577 individuals with a diagnosis and N≥3,277 individuals without a diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Five linear regression models will test the association between social media use (time- and engagement-based dimensions; a-e) with mental health diagnosis (two levels: *no diagnosis, diagnosis*), with the *no diagnosis* group set as the reference level. We will employ equivalence tests to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and mental health diagnosis follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with any mental health diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A positive and statistically significant association between social media use and mental health diagnosis will suggest that adolescents with any mental health diagnosis score higher on the examined dimension of social media use (a-e) compared to adolescents without a diagnosis. • In summary, Hypothesis 1.1 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and mental health diagnosis is statistically significant, and c) the regression coefficient reflecting this association is positive.

1) Investigate whether adolescents with any mental health condition use social media differently than those without a condition.	Hypothesis 1.2) Adolescents with any mental health diagnosis will score lower than adolescents without a diagnosis in f) happiness about the number of online friendships, g) honest online self-disclosure, and h) authentic self-presentation online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least N≥577 individuals with a diagnosis and N≥3,277 individuals without a diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Three linear regression models will test the association between each dimension of social media engagement (f-h) and mental health diagnosis (two levels: *no diagnosis, diagnosis*), with the *no diagnosis* group set as the reference level. We will employ equivalence tests to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and mental health diagnosis follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with any mental health diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A negative and statistically significant association between social media engagement and mental health diagnosis will suggest that adolescents with any mental health diagnosis score lower on the examined dimension of social media engagement (f-h) compared to adolescents without a diagnosis. • In summary, Hypothesis 1.2 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and mental health diagnosis is statistically significant, and c) the regression coefficient reflecting this association is negative.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.0) Adolescents with internalising mental health diagnoses will not differ from adolescents without a diagnosis in c) lack of control over time spent online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least N≥370 individuals with internalising only diagnoses and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	One linear regression model will test the association between social media engagement (c) and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *externalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and internalising diagnosis lies between the equivalence bounds, -0.4 and 0.4.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with an internalising diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A non-significant association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) will suggest that adolescents with internalising diagnoses do not differ from adolescents without a diagnosis on lack of control over time spent online (c). • In summary, Hypothesis 2.0 will be supported if a) the CIs for this association fall within the equivalence bounds, and b) the association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) is not statistically significant.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.0) Adolescents with externalising mental health diagnoses will not differ from adolescents without a diagnosis in b) online social comparison, d) monitoring of online feedback and e) feeling impacted by online feedback, g) honest online self-disclosure, and h) authentic self-presentation online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least N≥199 individuals with externalising only diagnoses and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Five linear regression models will test the association between social media use (b, d, e, g, h) and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *externalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with an externalising diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A non-significant association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) will suggest that adolescents with externalising diagnoses do not differ from adolescents without a diagnosis on the considered dimension of social media engagement (b, d, e, g, h). • In summary, Hypothesis 2.0 will be supported if a) the CIs for this association fall within the equivalence bounds, and b) the association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) is not statistically significant.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.1) Adolescents with internalising mental health diagnoses will score higher than adolescents without a diagnosis in a) time spent on social media, b) online social comparison, d) monitoring of online feedback and e) feeling impacted by online feedback.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥370 individuals with internalising only diagnoses and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Four linear regression models will test the association between social media use (time- and engagement-based dimensions; a, b, d, e) and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *internalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with an internalising diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A positive and statistically significant association between social media engagement and diagnostic category (comparing *internalising* and *no diagnosis*) will suggest that adolescents with an internalising diagnosis score higher on the examined dimension of social media engagement (a, b, d, e) compared to adolescents without a diagnosis. • In summary, Hypothesis 2.1 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) is statistically significant, and c) the regression coefficient reflecting this association is positive.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.2) Adolescents with internalising mental health diagnoses will score lower than adolescents without a diagnosis in f) happiness about online friendships, g) honest online self-disclosure and h) authentic self-presentation online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥370 individuals with internalising only diagnosis and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Three linear regression models will test the association between each dimension of social media engagement (f, g, h) and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *internalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with an internalising diagnosis do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A negative and statistically significant association between social media engagement and diagnostic category (comparing *internalising* and *no diagnosis*) will suggest that adolescents with an internalising diagnosis score lower on the examined dimension of social media engagement (f, g, h) compared to adolescents without a diagnosis. • In summary, Hypothesis 2.2 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *internalising* and *no diagnosis*) is statistically significant, and c) the regression coefficient reflecting this association is negative.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.3) Adolescents with externalising mental health diagnoses will score higher than adolescents without a diagnosis in a) time spent on social media and c) lack of control over time spent online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥199 individuals with externalising only diagnoses and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Two linear regression models will test the association between each dimension of social media use and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *externalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with externalising only diagnoses do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A positive and statistically significant association between social media engagement and diagnostic category (comparing *externalising* and *no diagnosis*) will suggest that adolescents with externalising only diagnoses score higher on the examined dimension of social media engagement (a, c) compared to adolescents without a diagnosis. • In summary, Hypothesis 2.3 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) is statistically significant, and c) the regression coefficient reflecting this association is positive.

2) Investigate whether adolescents with an internalising or externalising condition use social media differently than those without a condition.	Hypothesis 2.4) Adolescents with externalising diagnoses will score lower than adolescents without a diagnosis in f) happiness about the number of online friendships.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥199 individuals with externalising only diagnoses and *N*≥3,277 with no diagnosis. Results indicated 100% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 100% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	One linear regression model will test the association between happiness about the number of online friends and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *no diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *externalising* vs *no diagnosis*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with externalising only diagnoses do not have a meaningful difference in their social media use compared to adolescents without a diagnosis. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A negative and statistically significant association between social media engagement and diagnostic category (comparing *externalising* and *no diagnosis*) will suggest that adolescents with externalising only diagnoses score lower on the examined dimension of social media engagement (f) compared to adolescents without a diagnosis. • In summary, Hypothesis 2.4 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *externalising* and *no diagnosis*) is statistically significant, b) the regression coefficient reflecting this association is negative.

3) Investigate whether adolescents with an internalising condition use social media differently than those with an externalising condition.	Hypothesis 3.0) Adolescents with internalising diagnoses will not differ from adolescents with an externalising diagnoses in a) time spent on social media and f) happiness about the number of online friendships.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥370 individuals with internalising only and *N*≥199 with externalising only diagnoses. Results indicated 96% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 99.8% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Two linear regression models will test the association between each dimension of social media engagement and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *internalising diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *internalising* vs *externalising*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with externalising diagnoses do not have a meaningful difference in their social media use compared to adolescents with internalising diagnoses. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A non-significant association between social media use and diagnostic category (comparing *internalising* and *externalising*) will suggest that adolescents with internalising diagnoses do not differ from adolescents with externalising diagnoses on the considered dimension of social media engagement (a, f). • In summary, Hypothesis 3.0 will be supported if a) the CIs for this association fall within the equivalence bounds, and b) the association between social media use and diagnostic category (comparing *internalising* and *externalising*) is not statistically significant.

3) Investigate whether adolescents with an internalising condition use social media differently than those with an externalising condition.	Hypothesis 3.1) Adolescents with internalising diagnoses will score higher than adolescents with externalising diagnoses in b) online social comparison, d) monitoring of online feedback, and e) feeling impacted by online feedback.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥370 individuals with internalising only and *N*≥199 with externalising only diagnoses. Results indicated 96% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 99.8% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Three linear regression models will test the association between each dimension of social media engagement and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *externalising diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *internalising* vs *externalising*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *internalising* and *externalising*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with internalising only diagnoses do not have a meaningful difference in their social media use compared to adolescents with externalising only diagnoses. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A positive and statistically significant association between social media engagement and diagnostic category (comparing *internalising* and *externalising*) will suggest that adolescents with internalising only diagnoses score higher on the examined dimension of social media engagement (b, d, e) compared to adolescents with externalising only diagnoses. • In summary, Hypothesis 3.1 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *internalising* and *externalising*) is statistically significant, and c) the regression coefficient reflecting this association is positive.

3) Investigate whether adolescents with an internalising condition use social media differently than those with an externalising condition.	Hypothesis 3.2) Adolescents with internalising diagnoses will score lower than adolescents with externalising diagnoses in c) lack of control over time spent online, g) online honest self-disclosure and h) authentic self-presentation online.	We calculated power given a theoretical SESOI of d = 0.4, and a sample size of at least *N*≥370 individuals with internalising only and *N*≥199 with externalising only diagnoses. Results indicated 96% power to reject effects larger than the SESOI using the *'power_t_TOST'* function for equivalence testing and 99.8% power to detect the SESOI in the hypothesised direction using the *'pwr.t2n.test'* function in R.	Three linear regression models will test the association between each dimension of social media engagement and diagnostic category (three levels: *no diagnosis, internalising, externalising*), with the *externalising diagnosis* group set as the reference level. For this hypothesis, we will examine the regression coefficient for *internalising* vs *externalising*. We will employ an equivalence test to assess whether the 90% confidence interval (CI) for the association between social media use and mental health diagnosis lies between the equivalence bounds, -0.4 and 0.4. We will employ regression models to assess whether the association between social media use and diagnostic category (comparing *internalising* and *externalising*) follows the hypothesised direction.	• If the CI for this association is within the equivalence bounds, it will suggest that adolescents with internalising only diagnoses do not have a meaningful difference in their social media use compared to adolescents with externalising only diagnoses. On the contrary, if the CIs do not fall within the equivalence bounds, it will suggest that the association is of meaningful magnitude. • A negative and statistically significant association between social media engagement and diagnostic category (comparing *internalising* and *externalising*) will suggest that adolescents with internalising only diagnoses lower higher on the examined dimension of social media engagement (c, g, h) compared to adolescents with externalising only diagnoses. • In summary, Hypothesis 3.2 will be supported if a) the CIs for this association do not fall within the equivalence bounds, b) the association between social media use and diagnostic category (comparing *internalising* and *externalising*) is statistically significant, b) the regression coefficient reflecting this association is negative.

Table 2. Summary of our categorisation of mental health diagnoses into internalising and externalising conditions.

Grouped diagnostic category	Diagnoses received in the DAWBA
Internalising	Separation anxiety disorder, generalised anxiety disorder, obsessive-compulsive disorder, specific phobia, social phobia, agoraphobia, panic disorder, post-traumatic stress disorder, other anxiety disorder, body dysmorphic disorder, major depressive episode, other depressive episode, and eating disorders.
Externalising	Hyperkinetic disorder, other hyperactivity disorder, oppositional defiant disorder, conduct disorder confined to family, unsocialised conduct disorder, socialised conduct disorder, other conduct disorders.
Excluded[a] from analyses in Question 2 and 3	Autism spectrum disorder, tic disorder, psychotic disorders, mania, and bipolar affective disorder.

Note [a]. We exclude adolescents diagnosed with these conditions as they do not clearly map onto the symptomatology of either internalising or externalising diagnoses[35,66]. Further, because the diagnostic variance of mania and bipolar disorder accounted for by internalising pathology is lower than most other internalizing disorders, such as anxiety and depression[33], we decided to exclude these diagnoses from the internalising category, together with autism spectrum disorder, tic disorder and psychotic disorders.

Table 3. Review of key social media and mental health literature used to formulate the hypotheses of our study.

Social media use	Scientific literature	Relevant hypotheses	Key references
Time spent on social media Item: "When you use social media sites or apps how much time in total do you spend using them on a typical school day/ weekend?"	Research on the relationship between social media use and mental health outcomes in different samples has yielded limited and conflicting results. On one hand, studies have found that young individuals diagnosed with depression tend to spend more time on social media compared to healthy controls[99]. Further, studies that focus on excessive rather than average time spent on social media show comorbidity between anxiety, depression, attention deficit hyperactivity disorder and excessive time spent on social media[73]. However, in a sample of hospitalized adolescents with psychiatric conditions, the frequency of social media use and perception of overuse was not associated with clinical severity[32]. Additionally, in an independent clinical sample, using social media less, both overall and for messaging, was linked to higher levels of suicidal ideation over the next 30 days[100]. There is more work on community samples available, examining the average time spent on social media (mostly self-reported) and its relation to depression, anxiety and other indicators of mental health. Recent reviews and meta-analyses have reached a general agreement that associations are weak and positive (higher social media use is linked with higher levels of anxiety and depression[8,15,101]). Overall, despite mixed evidence, research, therefore, seems to suggest a small relationship between more time spent on social media and lower mental health, both when considering internalising and externalising symptoms. We hypothesise this to hold for clinical samples that are, therefore, expected to spend more time on social media.	H1.1a: Adolescents with any mental health diagnosis will spend more time on social media than adolescents without a diagnosis. H2.1a: Adolescents with internalising mental health diagnoses will spend more time on social media than adolescents without a diagnosis. H2.3a: Adolescents with externalising mental health diagnoses will spend more time on social media than adolescents without a diagnosis. H3.0a: Adolescents with externalising mental health diagnoses will not differ from adolescents with internalising diagnoses in time on social media.	Valkenburg et al. (2022) Cunningham et al. (2017) George et al. (2017) Akkın Gürbüz et al. (2017) Hussain & Griffiths (2018) Hamilton et al. (2021) Riehm (2019)

Online social comparison Item: "I compare myself to others on social media sites and apps"	In offline contexts, social comparison provides means of gathering information about the social world. This is particularly important during adolescence when young people need to develop a sense of personal and social identity and adjust to bodily changes[44]. Despite social comparison being instrumental for maturation into adulthood, its exacerbation has been associated with poor mental health, particularly in relation to internalising symptoms. For instance, adolescents with anxiety and eating disorders engage in less favourable and more frequent social comparisons than adolescents without these conditions[45–49]. Social media platforms offer continuous and more concrete opportunities for social comparison than offline contexts, for instance, by allowing people to browse others' profiles without initiating social interaction[53]. Indeed, adolescents with a depression diagnosis reported unfavourably comparing themselves to others when using social media[30]. Similarly, among adolescents hospitalised for suicidal behaviour, 30% reported engaging in self-denigrating social comparisons, particularly body-related ones[31].	H1.1b: Adolescents with any mental health diagnosis will engage in more online social comparison than adolescents without a diagnosis. H2.0b: Adolescents with externalising mental health diagnoses will not differ in online social comparison from adolescents without a diagnosis. H2.1b: Adolescents with internalising mental health diagnoses engage in more online social comparison than adolescents without a diagnosis. H3.1b: Adolescents with internalising mental health diagnoses will engage in more online social comparison than adolescents with externalising diagnoses.	Krayer et al. (2007) Corning et al. (2006) Goodman et al. (2019) Rao et al. (2007) Thwaites & Dagnan, (2004) Troop et al. (2016) Pempek et al. (2009) Radovic et al. (2017) Weinstein et al. (2021)

Perceived lack of control over time spent online Item: "In general, I spend more time on social media than I mean to"	Difficulties in managing personal goal pursuit in the face of internal, interpersonal, and environmental forces that would impede it have been closely linked to the symptomatology of different mental health conditions, particularly externalising ones. For example, ADHD and conduct problems are characterised by lower levels of self-monitoring, self-instruction and goal setting[54]. Technological developments now allow individuals to access social media at any location or time of the day. Consequently, online platforms are not limited to a particular environment and repetitions of certain behaviors, such as opening an app or scrolling through one's feed[102], might result in adolescents feeling unable to reduce the time they spend online despite being motivated to do so. We expect difficulties in self-regulation and goal pursuit to also reflect in how adolescents with mental health conditions engage with social media. Indeed, in a sample of adolescent psychiatric inpatients, self-regulation of social media behaviour was associated with mental health symptoms[103]. Further, in qualitative interviews, 40% of inpatients hospitalised for suicidal behaviour reported trouble regulating social media use and feeling "addicted".[31]	H1.1c: Adolescents with any mental health diagnosis will be more likely to lack control over time spent online than adolescents without a diagnosis. H2.0c: Adolescents with internalising mental health diagnoses will not differ in lack control over time spent online from adolescents without a diagnosis. H2.3c: Adolescents with externalising mental health diagnoses will be more likely to lack control over time spent online than adolescents without a diagnosis. H3.2c: Adolescents with externalising mental health diagnoses will be more likely to lack control over time spent online than adolescents with internalising diagnoses.	Strauman et al. (2017) Winds et al. (2022) Weinstein et al. (2021) Bayer et al. (2022)

Monitoring of online feedback Item: "I monitor the amount of likes, comments and shares I get on social media"	Adolescents and adults with depression, anxiety and eating disorders, among other internalising conditions, have a higher tendency to seek feedback from others[104], particularly when negative, and engage in more reassurance seeking[56]. Further, compulsivity in social media checking behaviours has been associated with anxiety, depression and problematic smartphone use[105], therefore, becoming a cause of clinical and developmental concern[77]. Consistent with these results, online feedback-seeking has been associated with depressive symptoms in adolescent community samples[55]. This association holds even after accounting for the effects of the overall frequency of technology use, offline excessive reassurance-seeking, and prior depressive symptoms.	H1.1d: Adolescents with any mental health diagnosis will be more likely to monitor online feedback than adolescents without a diagnosis. H2.0d: Adolescents with externalising mental health diagnoses will not differ in monitoring online feedback from adolescents without a diagnosis. H2.1d: Adolescents with internalising mental health diagnoses will be more likely to monitor online feedback than adolescents without a diagnosis. H3.1d: Adolescents with internalising mental health diagnoses will be more likely to monitor online feedback than adolescents with externalising diagnoses.	Hames et al. (2013) Gillet et al., (2018) Elhai et al. (2020) Barry et al. (2017) Nesi & Prinstein (2015)

Perceived impact of online feedback Item: "The amount of likes, comments and shares I get on social media has an impact on my mood"	Compared to people without a mental health condition, individuals with internalizing conditions (e.g., depression[106]) tend to have biased perceptions and recall of interpersonal feedback. This bias is apparent in both their symptoms, such as negative beliefs about the self and worry, as well as neural markers, such as feedback- and error-related negativity[57,58]. Social media platforms offer continuous opportunities for exposure to feedback from peers and strangers. Further, feedback is often more quantifiable and permanent than when received offline (e.g., the number of likes received, and comments to old posts can be revisited). We, therefore, expect the biased perception of social feedback demonstrated in clinical groups to occur and possibly be heightened in online contexts[55].	H1.1e: Adolescents with any mental health diagnosis will be more likely to feel impacted by online feedback than adolescents without a diagnosis. H2.0e: Adolescents with externalising mental health diagnoses will not differ in feeling impacted by online feedback from adolescents without a diagnosis. H2.1e: Adolescents with internalising mental health diagnoses will be more likely to feel impacted by online feedback than adolescents without a diagnosis. H3.1e: Adolescents with internalising mental health diagnoses will be more likely to feel impacted by online feedback than adolescents with externalising diagnoses.	Gotlib (1983) Tobias & Ito (2021) Tucker et al. (2003) Nesi and Prinsten (2015)

Online friendship Item: "I am happy with the number of friends I have on social media"	In offline contexts, social connections are a protective factor against long-term adverse physical and emotional outcomes, particularly during adolescence[107]. In line with this evidence, young people with both internalising and externalising mental health conditions report difficulties with peers, having few friends and wanting to be alone[59–61]. Further, adolescents with internalising conditions tend to be unhappy with their social status and adolescents with externalising disorders tend to have trouble making and keeping friends, both online and offline[74,76].	H1.2f: Adolescents with any mental health diagnosis will be less happy about the number of online friends they have than adolescents without a diagnosis. H2.2f: Adolescents with internalising mental health diagnoses will be less happy about the number of online friends they have than adolescents without a diagnosis. H2.4f: Adolescents with externalising mental health diagnoses will be less happy about the number of online friends they have than adolescents without a diagnosis. H3.0f: Adolescents with internalising mental health diagnoses will be as happy about the number of online friends they have as adolescents with externalising diagnoses.	Viner et al. (2012) Asselmann et al. (2018) McBride & Preyde (2022) Sibley et al. (2010) Bagwell et al. (2001) Dawson et al. (2019)

Online self-disclosure Item: "I can be honest with people on social media sites and apps about how I am feeling"	Self-disclosure is a communication process by which one person reveals information about themselves to another[108]. The extent of self-disclosure has been associated with higher relationship quality, intimacy and well-being in offline contexts. In adolescent psychiatric inpatients, low levels of self-disclosure have been linked to suicidality, with anxiety and depression mediating this association[62]. The quality of self-disclosure also differs in people with low compared to high psychological distress. For instance, people that are less distressed tend to disclose more positive information whereas those high in distress tend to disclose more negative and less honest information[63]. This effect appears to also occur on social media platforms. For instance, depressed individuals are more likely to post darker, bluer and greyer images than people without depressive symptoms[109]. Communicating personal and emotional information increases the risk of embarrassment and rejection[110], which is compounded by adolescents' increased sensitivity to peer feedback and anxiety regarding negative social evaluations[111]. We expect this to be especially difficult for adolescents with internalising disorders, as they must balance the rewards associated with self-disclosure[112] with considerations of how that disclosure might be received by their peers.	H1.2g: Adolescents with any mental health diagnosis will engage in less online self-disclosure than adolescents without a diagnosis. H2.0g: Adolescents with externalising mental health diagnoses will not differ in online self-disclosure from adolescents without a diagnosis. H2.2g: Adolescents with internalising mental health diagnoses will engage in less online self-disclosure than adolescents without a diagnosis. H3.2g: Adolescents with internalising mental health diagnoses will engage in less online self-disclosure than adolescents with externalising diagnoses.	Sprecher & Hendrick (2004) Horesh & Apter (2006) Chen, H. (2017) Reece & Danforth (2017) Omarzu (2000) Van den bos (2014)

Authentic online self-presentation Item: "My social media profile is a true reflection of myself"	Different internalising conditions such as anxiety, depression and eating disorders are characterised by the internalisation of an ideal self that, once compared to perceptions of the actual self, results in negative self-evaluations. Adolescents with these conditions, therefore, create perfectionistic self-presentations to combat negative self-narratives and project desirable images of themselves in the mind of others[64,65,69–71]. We expect this process to also occur online, given the multiple affordances offered by social media platforms to curate one's image, such as deleting old posts, editing new posts and optimising messages before sending. Hence, adolescents with a mental health condition, particularly if an internalising condition, might be more likely to engage in impression management online to compensate for negative self-evaluations[63].	H1.2h: Adolescents with any mental health diagnosis will engage in less authentic online self-presentation than adolescents without a diagnosis. H2.0h: Adolescents with externalising mental health diagnoses will not differ in authentic online self-presentation from adolescents without a diagnosis. H2.2h: Adolescents with internalising mental health diagnoses will engage in less authentic online self-presentation than adolescents without a diagnosis. H3.2h: Adolescents with internalising mental health diagnoses will engage in less authentic online self-presentation than adolescents with externalising diagnoses.	Flett & Hewitt (2014) Jain & Sudhir (2010) O'Connor et al. (2010) Sassaroli et al. (2008) Calvo et al. (2009) Chen, H. (2017)

Supplementary Methods

Social media variables

The questions capturing social media engagement were developed through consultation with a youth advisory board conducted by the National Center of Social Research in 2015. The board included 10 young people aged 16-24 years old, that were identified through existing mental health research networks and came from across England, with a wide range of experiences and interests. They all shared a common interest in mental health, linked to either first-lived experiences or involvement in the field.

Due to routing (MHCYP, 2017), the SMTimeSpent question (weekday and weekend; SMTimeSpentS/SMTimeSpentw) were only asked of adolescents who responded to another survey question that they use social media (SMUse) and the frequency of use (SMFreqofUse) is "daily or most days". Adolescents that responded that they used social media sites between "a few times a week" to "less often than once a month" did not complete the SMTimeSpentS/W questions. Supplementary Table 1 provides an overview of the routing for the relevant social media questions. Supplementary Table 2 details our approach to handling missing data for the SMTimeSpent variables.

Supplementary Table 1. Items in the MHCYP survey used to measure time spent on social media.

Variable	Use in analysis	Label	Answers	Routing
SMUse	No	Do you use any of the following social media sites or apps?	1 = AskFM; 2 = Bebo; 3 = Blogger; 4 = Facebook; 5 = Flickr; 6 = Google Hangouts and/or Google Plus (Google+); 7= Hi5; 8 = Instagram; 9 = Jabble; 10 = MySpace; 11= Piczo; 12 = Pinterest; 13 = SnapChat; 14 = Tumblr; 15 = Twitter; 16 = Vimeo; 17 =Vine; 18 = WhatsApp; 19 = YouTube; 20 = Skype; 21 = Musical.ly; 22 = Other; 23 =I do not use social media apps or sites.	None

SMFreqofUse	No	How often do you use social media sites or apps?	1 = Daily or most days; 2 = A few times a week; 3 = Once a week; 4 = A few times a month; 5 = Once a month; 6= Less often than once a month.	SMUse = 1 to 20, 22
SMTimeSpentS	Yes	When you use social media sites or apps how much time in total do you spend using them on a typical school day?	1 = Less than 30 minutes; 2 = More than 30 minutes but less than an hour; 3 = One to two hours 4 = Two to three hours; 5 = Three to four hours; 6 = Four to five hours; 7 = Five to six hours 8 = Six to seven hours; 9 = More than seven hours.	SMFreqofUse = 1
SMTimeSpentW	Yes	When you use social media sites or apps how much time in total do you spend using them on a typical weekend or holiday day?	1 = Less than 30 minutes; 2 = More than 30 minutes but less than an hour; 3 = One to two hours 4 = Two to three hours; 5 = Three to four hours; 6 = Four to five hours; 7 = Five to six hours 8 = Six to seven hours; 9 = More than seven hours.	SMFreqofUse = 1

Supplementary Table 2. Planned approach to missing data for the time spent items in the MHCYP survey.

Variables	Missing values	Approach to missing data
SMTimeSpent (SMTimeSpentS and SMTimeSpentW)	Adolescents that responded to SMFreqofUse with anything other than 1 (Daily or most days) have missing values for SMTimeSpentS and SMTimeSpentW.	For people that answered 2-4 on SMFreqUse (2 = A few times a week; 3 = Once a week; 4 = A few times a month), we will input 30 minutes on the *SMTimeSpent* item. For people that answered 5-6 on SMFreqofUse (5 = Once a month; 6 = Less often than once a month), we will input zero minutes on the *SMTimeSpent* item.

Mental health variables

During the interview, participants completed the Strength and Difficulties Questionnaire (SDQ) and the Development and Wellbeing Assessment (DAWBA). The SDQ was administered first (Supplementary Table 4). This tool was used to increase the validity of the diagnostic assessment by allowing the identification of participants with mental health symptoms not picked up by the initial screening items on the DAWBA. Hence, if a participant scored highly on the SDQ, the interviewer was directed to ask in-depth questions from the DAWBA internalising modules.

The DAWBA uses structured and semi-structured questions to assess the presence and severity of symptoms for a wide range of DSM-5 or ICD-10 mental health disorders. Each disorder is assigned a separate module. Each module starts with a few screening items (see Supplementary Table 4; e.g., “In the last 4 weeks, have there been times when you have been very sad, miserable, unhappy or tearful?”). If a participant responds positively to these structured items, they are subsequently asked unstructured questions about these problems (e.g., “Please describe your mood – sadness or irritability - and your level of interest in things”).

During the assessment, interviewers transcribed unstructured responses on the DAWBA verbatim and were also able to add personal comments beneath each response. If a participant did not endorse one of the structured items (i.e., did not report mental health symptoms), the interview proceeded to the next module. Having collected the information from respondents, the DAWBA online system then processes and presents the information to make it as easy as possible for trained clinical raters to decide if the child has a psychiatric disorder. In Supplementary Figure 1, we provide a fictitious exemplar of a DAWBA module as shown to the clinical raters. Clinical raters can assign DSM-5 or ICD-10 mental health diagnoses based on the collected information. In Supplementary Figure 2, we provide a fictitious exemplar of the DAWBA overview completed by the clinical rater to assess the presence of a disorder.

Supplementary Table 3. Items from the Strength and Difficulties Questionnaire.

Item	Answer options		
I try to be nice to other people. I care about their feelings.	Not true	Somewhat true	Certainly true
I am restless, I cannot stay still for long.	Not true	Somewhat true	Certainly true
I get a lot of headaches, stomach-aches or sickness.	Not true	Somewhat true	Certainly true
I usually share with others (food, games, pens etc.).	Not true	Somewhat true	Certainly true
I get very angry and often lose my temper	Not true	Somewhat true	Certainly true
I am usually on my own. I generally play alone or keep to myself.	Not true	Somewhat true	Certainly true
I usually do as I am told.	Not true	Somewhat true	Certainly true
I worry a lot.	Not true	Somewhat true	Certainly true
I am helpful if someone is hurt, upset or feeling ill.	Not true	Somewhat true	Certainly true
I am constantly fidgeting or squirming.	Not true	Somewhat true	Certainly true
My attention is good.	Not true	Somewhat true	Certainly true
I have one good friend or more.	Not true	Somewhat true	Certainly true
I fight a lot. I can make other people do what I want.	Not true	Somewhat true	Certainly true
I am often unhappy, down-hearted or tearful.	Not true	Somewhat true	Certainly true

Other people my age generally like me	Not true	Somewhat true	Certainly true
I am easily distracted, I find it difficult to concentrate.	Not true	Somewhat true	Certainly true
I am nervous in new situations. I easily lose confidence.	Not true	Somewhat true	Certainly true
I am kind to younger children.	Not true	Somewhat true	Certainly true
I am often accused of lying or cheating.	Not true	Somewhat true	Certainly true
Other children or young people pick on me or bully me.	Not true	Somewhat true	Certainly true
I often volunteer to help others (parents, teachers, children).	Not true	Somewhat true	Certainly true
I think before I do things.	Not true	Somewhat true	Certainly true
I take things that are not mine from home, school or elsewhere.	Not true	Somewhat true	Certainly true
I get on better with adults than with people my own age.	Not true	Somewhat true	Certainly true
I have many fears, I am easily scared.	Not true	Somewhat true	Certainly true
I finish the work I'm doing.	Not true	Somewhat true	Certainly true

29 **Supplementary Table 4.** Example of structured question from a DAWBA module on attention hyperactivity disorder.

DAWBA module	Example questions
Attention deficit hyperactivity disorder	11-19: Whether people outside the family complain about you having problems with overactivity or poor concentration?
	11-19: Whether your family complain about you having problems with overactivity or poor concentration
	11-19: Whether respondent thinks they have real problems with overactivity or poor concentration
	11-19: Thinking about the last 6 months: Do you often fidget?
	11-19: Thinking about the last 6 months: Is it hard for you to stay sitting down for long?
	11-19: Thinking about the last 6 months: Are you too full of energy and always on the go?
	11-19: Thinking about the last 6 months: Do you find it difficult to be quiet?
	11-19: Thinking about the last 6 months: Is it hard for you to unwind and relax?
	11-19: Thinking about the last 6 months: Do you often blurt out an answer before you have heard the question properly, or finish other people s sentences for them?
	11-19: Thinking about the last 6 months: Is it hard for you to wait your turn?
	11-19: Thinking about the last 6 months: Do you often interrupt other people when they are busy?
	11-19: Thinking about the last 6 months: Do you often go on talking even if you have been asked to stop, or if no one is listening?
	11-19: Thinking about the last 6 months: Do you often make careless mistakes or fail to pay attention to what you are supposed to be doing?
	11-19: Thinking about the last 6 months: Do you often lose interest in what you are doing?

11-19: Thinking about the last 6 months: Do you often find it hard to concentrate on what other people are saying to you?
11-19: Thinking about the last 6 months: Is it often hard for you finish a job properly, doing all the last bits after the most interesting bits have been done?
11-19: Thinking about the last 6 months: Is it often hard for you to get yourself organized to do something?
11-19: Thinking about the last 6 months: Do you often try to get out of things you would have to think hard about?
11-19: Thinking about the last 6 months: Do you often lose things you need for study or work?
11-19: Thinking about the last 6 months: Are you easily distracted?
11-19: Thinking about the last 6 months: Are you often forgetful?
11-19: Whether concerned at present about any aspect of concentration or activity level?
11-19: How much have difficulties with activity or concentration upset or distressed you?
11-19: Have difficulties with attention or activity level interfered with: How well you get along with the people you are closest to (e.g. family, partner)?
11-19: Have difficulties with attention or activity level interfered with: Making and keeping friends?
11-19: Have difficulties with attention or activity level interfered with: Work or study?
11-19: Have difficulties with attention or activity level interfered with: Hobbies, sports or other leisure activities?
11-19: Have difficulties with attention or activity level made it harder for those around you (family, friends, etc.)?

Supplementary Figure 1.

Social Phobia **youth*in*mind**

Dawba ID: 193082 DSM-IV prediction: ++

Details: 15 years old, Male ICD-10 prediction: ++

	Parent	Self
Any concerns?	Yes	Yes
Anxious about:		
Meeting new people	++	++
Meeting a lot of people	+	++
Eating in front of others	-	-
Speaking in class	+	+
Reading aloud in front of others	++	++
Writing in front of others	+	+
Separation or social anxiety?	Social	Social
Frightened with adults/kids	Adults and kids	Adults and kids
Can socialise with familiar people	++	++
Due to fear of embarrassment	++	++
Due to delay in speech, writing, reading	-	+
Age of onset	?	?
Duration in months	6+	6+
Blushes in social situations		-
Feels sick in social situations		-
Urgency in social situations		-
Upset when social fear is triggered	++	++
How often social fear is triggered	Rarely	Most weeks
Avoids relevant social situations	++	++
Avoidance interferes with daily life	+	++
Child thinks fear is excessive	++	+
Child upset to have social fears	++	++
Burden	+	+

>> Open-ended comments: Parent

Description of the problem
Having to speak to new people.

How often?
About once a week.

How severe?
To the point where he will actively avoid people.

Interfering with quality of life?
It's making him reluctant to leave the house.

Social Phobia - 1 Created on: 12/07/2016 10:43

Social Phobia **youth*in*mind**

Dawba ID: 193082 DSM-IV prediction: ++

Details: 15 years old, Male ICD-10 prediction: ++

Done anything about it?
Encouraged him to start doing some kind of activity.

>> Open-ended comments: Self

Description of the problem
Having to speak to people my age outside of my close friend group.

How often?
About once a day.

How severe?
To the point where I have to actively avoid social interaction by walking around people.

Interfering with quality of life?
It makes me hesitant to leave the house.

Done anything about it?
I've found a community in the online image board 4chan, specifically /r9k/ which has helped knowing that other people are in a similar situation.

Social Phobia - 2 Created on: 12/07/2016 10:43

A fictional example of a participant DAWBA module used to showcase information collected during the interview as seen by the clinical rater.

50 **Supplementary Figure 2.**

Overview of Development and Well-Being Assessment **youth*in*mind**

Dawba ID: 193082

Details: 15 years old, Male

INFORMANTS

ID	Informant	Date	Age	Gender	About
193082	Parent	12/07/2016	15	Male	Tim
193082	Self	12/07/2016	15	Male	

SYMPTOMS, IMPACT AND DIAGNOSTIC PREDICTIONS

Predicted probability of disorder: high

	Parent		Self		Prediction	
	Symp	Imp	Symp	Imp	DSM	ICD
Autism Spectrum Disorder	++	+++			++	++
Separation Anxiety	-		-		- -	- -
Specific Phobia	+		+	-	-	-
Social Phobia	++	++	++	+++	++	++
Panic	-		++	+++	-	-
Agoraphobia	-		-		-/+	-/+
PTSD	-		-		- -	- -
OCD	-		+		-	-
BDD						
Generalised Anxiety	-		-		- -	- -
Depression	++	-	+++	+++	++	++
Deliberate Self Harm	-		-			
Bipolar						
Emotions at school						
Attachment difficulties						
Hyperactivity	-				- -	- -
Oppositional	-				-/+	-/+
Conduct	-		+		- -	- -
Anorexia / Bulimia	-		-		- -	- -
Tics	-		-		- -	- -
Psychosis						
Feeding difficulties						
Sleeping difficulties						
Elimination difficulties						
Other concerns	-		-			

CLINICAL DIAGNOSES ASSIGNED BY RATER

DSM-IV diagnoses

Asperger

Rated by: 12/07/2016

ICD-10 diagnoses

Overview of Development and Well-Being Assessment - 1 Created on: 12/07/2016 10:42

51

52 A fictional example of a participant DAWBA used to showcase information provided by the clinical rater to allocate mental health diagnoses after reviewing
53 each DAWBA module.

Exploratory analyses for specific diagnoses or diagnostic categories

The Registered Report will include exploratory analyses on specific diagnoses, as reported below. Of note, we decided to include these analyses as exploratory because a) our ability to conduct them will be conditioned on sample size and power, and b) we do not have specific hypotheses related to these questions.

To increase the clinical utility of this Registered Report we will run additional exploratory analyses examining responses to all social media questions (a-g) separately for individual diagnoses (e.g., major depressive disorder) or diagnostic categories (e.g., depressive disorders, anxiety disorders). First, we will present descriptive statistics for each specific diagnosis, as long as there are at least 10 individuals with that diagnosis. Next, for individual diagnoses or diagnostic categories with sufficient sample size, we will conduct follow up analyses to test whether individuals falling within each group respond differently to each social media question (a-g) than individuals with no diagnosis.

As comorbid mental health conditions are extremely common, we will include individuals with multiple conditions in these analyses. Further, we will conduct sensitivity analyses to assess whether the results are meaningfully different when individuals with comorbidities are removed.

Choice of the Smallest Effect Size of Interest

We chose to identify our smallest effect sizes of interest (SESOI) at different levels of explanation (i.e., theoretically, practically and clinically meaningful). Given the aim of our study to test the differences in social media use across clinical and non-clinical populations, we believe that no single SESOI provides a comprehensive answer as to what is a meaningful effect. Our research questions hold significance for a diverse range of stakeholders, including researchers, policymakers, the general public and clinicians, each requiring insights from a different level of explanation. For instance, with a theoretically meaningful SEOSI as the benchmark, our findings can be contextualised in relation to the broader field of mental health research. On the contrary, with practically and clinically meaningful SESOIs, our results can aid decision-making for policy and clinical practice.

Therefore, while a theoretically meaningful SESOI based on everyday behaviours and their link to mental health will serve as our primary SESOI for the power calculations and equivalence tests (to allow for a clear confirmatory approach), secondary SESOIs will play an important role in supporting the interpretation of our findings, which we plan to specifically address in a separate section of the Stage 2 results. Overall, this will ensure the applicability of our study to both academic and practical domains.

Primary SESOI

A theoretically meaningful SESOI serves as a way to assess the significance of a specific effect size in the context of a research question, using a theoretical framework as the reference point. After an extensive scoping exercise to identify a suitable theoretical foundation, we determined that the most relevant benchmarks for our research questions are everyday behaviours linked to mental health, such as sleep and physical activity. These behaviours, much like social media use, are a regular part of daily routines but, unlike the latter, they are well-established markers of psychopathology based on both theory (Crone et al. 2005; Rebar et al., 2017; Knowlden et al., 2012) and matched empirical evidence (Mollayeva et al., 2016; Ruscio et al., 2008). Consequently, if the effect size of social media use is comparable to that observed for behaviours like sleep and physical activity, we can conclude that social media use also represents an everyday behaviour that exhibits meaningful group-level differences between clinical and non-clinical populations.

An alternative possibility that we considered to identify the theoretical SESOI is the use of social media literature. However, despite the small effect sizes that have made headlines and attracted public attention in this space (Burn-Murdoch, 2023; Haidt, 2023), there is still a lack of theoretical foundation

as to what effects can be considered meaningful. This combined with a lack of studies that a) directly compare social media use in clinical vs non-clinical populations (see Fassi et al., 2023 for a review) and b) use comparable measures of engagement to the ones examined in our study, makes it currently impossible to adequately identify the theoretical SESOI of interest solely based on social media research. We therefore decided to draw from research on everyday behaviours other than social media use, as expanded on below.

In the case of sleep, research suggests a meta-analytic effect size of WMD = .48 (i.e., weighted mean difference), corresponding to d = 0.413 (Ruscio, 2008), for the difference between clinical and non-clinical populations in sleep duration (Mollayeva et al., 2016). In terms of physical exercise, a large (N = 78,886) nationally representative study (Okoro et al., 2005) suggests that, when compared to healthy individuals, individuals with mental health conditions are less likely to engage in recommended levels of physical activity, with a prevalence ratio = 0.84 after controlling for other health factors, corresponding to a medium effect size. Based on this evidence, we define d = 0.4 as the smallest theoretically meaningful effect size that would indicate a difference in social media use between clinical and non-clinical populations at least as large as that found in well-defined behavioural markers of psychopathology. We use this as our primary SESOI, informing both power calculations and equivalence tests.

Secondary SESOIs

Given the relevance of our research question for different stakeholders (e.g., researchers, the public/policy sector, and clinicians), we also chose to identify SESOIs based on practical and clinical meaningfulness. This information will aid the interpretation of our findings that we plan to address in a separate section of the Stage 2 results named “Results into context: practical and clinical SESOI”.

Practically meaningful. A practically meaningful SESOI represents the level of change in a phenomenon that individuals are likely to subjectively notice. Setting a benchmark for practical significance helps researchers define thresholds for meaningful differences or effects that are relevant in real-world scenarios. Reviews of the literature suggest that, across various physical health conditions including pain, cardiovascular and respiratory disease, when patients are asked to subjectively identify a minimal change in symptomatology, their estimates consistently fall close to half a standard deviation, corresponding to d = 0.5 (Norman et al., 2003). This value approximates the range of human discrimination first identified by Miller (1956), who characterised the limit of people's abilities across a wide range of discrimination tasks as being equivalent to d = 0.36-0.63. Based on these grounds, d = 0.5 has been employed as a practically meaningful anchor of change in research on experimental

design for media effects (Ferguson, 2009; that used d = .41) and screen time (Przybylski et al., 2020). Therefore, we set d = 0.5 as the benchmark for the smallest practically meaningful effect size (i.e., a difference in social media use that an individual might subjectively notice).

Clinically meaningful. An effect is clinically meaningful to the extent that it can distinguish between clinical and non-clinical populations. For example, psychotherapy research suggests that clinically meaningful interventions are those that reduce psychological distress such that an individual with a mental health diagnosis - and therefore high levels of distress - will report non-clinical levels of distress following treatment (Jacobson & Truax, 1992). To quantify the magnitude of change that is clinically meaningful, previous research has examined Health-Related Quality of Life (HRQoL) as a transdiagnostic index of distress across clinical and non-clinical populations (Guyatt, 1993). The HRQol is a multi-dimensional measure that captures both the physical and mental health status of individuals as well as the overall impact that health status has on their quality of life. Studies consistently show that HRQoL differs between patients with and without mental health conditions. For instance, one meta-analysis found that patients with generalised anxiety disorder differed from controls by d = 1.35 in HRQoL (Olatunji et al., 2007). Similarly, studies of children and adolescents diagnosed with ADHD have found differences in HRQoL of d = .72 and d = 1.05 compared to controls (Hampel & Desman, 2006; Varni & Burwinkle, 2006). Given that reasonable SESOIs based on clinical significance are all larger than our established theoretical SESOI (d = 0.4), with a range of d = 0.72-1.35, our study will be powered to detect clinically meaningful differences in social media use across clinical and non-clinical populations.

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